Human-Mediated Sensing for AI Decision-Making under Uncertainty

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Abstract

The performance of AI agents in organizations and infrastructure is increasingly constrained not by computation, but by limited access to information not captured in databases or APIs. Humans already act as implicit sensors for agent by answering queries, performing tests, and reporting observations; however, existing systems treat this input as ad hoc rather than as a resource that can be explicitly modeled and optimized. We propose a framework that formalizes humans and human-mediated tests as explicit *sensing actions*, or a "human API" (H-API) for agentic systems. Our framework captures how human input reduces uncertainty, shapes downstream decisions, and depends on when, how, and whether people provide information, all while preserving human control over information flow. Unlike prior approaches (e.g., Value of Information, Bayesian experimental design, Active Feature Acquisition) that treat queried data as fixed and reliably generated, H-API considers human input to be a variable and strategic resource, whose availability, reliability, and incentives must be explicitly accounted for. Applications include clinical decision support in healthcare settings, where additional input can degrade performance and where selectively withholding information access may lead to improved outcomes. Our source code is available in the Supplementary Material.

1 Introduction

Modern AI agents increasingly operate through structured interfaces such as tools, APIs, and programmatic workflows. In customer support systems, agents query ticketing systems and CRMs [Bubeck et al., 2023]; in clinical settings, diagnostic assistants retrieve and reason over electronic health records [Topol, 2019, Esteva et al., 2019]; and in planning and logistics domains, agents compose sequences of tool calls for scheduling and routing [Russell and Norvig, 2021]. In these settings, the agent's interaction with the world is fully mediated by a finite set of machine-readable observations. Despite this structured access, many decision-relevant variables remain systematically outside available interfaces. These include physical state variables (*e.g.*, "is there standing water at this bridge?") not captured by sensors, tacit institutional knowledge (*e.g.*, "is this project politically sensitive?"), and context-dependent social norms (*e.g.*, "would this intervention be considered appropriate here?"). Consequently, real-world systems frequently rely on humans to supply missing information through queries, clarification, or intervention [Amershi et al., 2014, Bhatt et al., 2023].

A large body of prior work has studied information acquisition under uncertainty, including value-of-information (VOI) theory [Lindley, 1956, Wilson, 2015], Bayesian experimental design [Cheng and Shen, 2005, Zhou et al., 2008, Cavagnaro et al., 2010, Houlsby et al., 2011, Hennig and Schuler, 2012, Myung et al., 2013, Shahriari et al., 2015, Hernández-Lobato et al., 2016, Wang and Jegelka, 2017, Vincent and Rainforth, 2017, Gal et al., 2017], active feature acquisition [Krause et al., 2008, Settles, 2010], and partially observable Markov decision processes (POMDPs) [Kaelbling et al., 1998]. These frameworks provide a building block for deciding when additional observations are worth their cost. However, these approaches rely on a key modelling assumption that limits their applicability to modern agentic systems: *the set of observations is assumed to be pre-specified, engineered, and directly machine-executable*. In classical settings, sensing actions correspond to well-defined variables (e.g., acquiring a feature, running a test, or querying a known simulator), and the action space is typically fixed and fully instrumented. In contrast, modern AI systems operate in environments where a substantial portion of informative signals is only accessible through human-mediated, heterogeneous, and non-uniform interaction channels. These include asking questions to different people, requesting different types of judgment, or triggering institutional processes whose outputs are not standardized, repeatable, or fully reliable. This creates a structural gap which remains fully underexplored: *existing formulations do not directly model human interaction as a compositional, multi-channel, cost-structured sensing interface*, which is the dominant mode of information acquisition in deployed agentic systems.

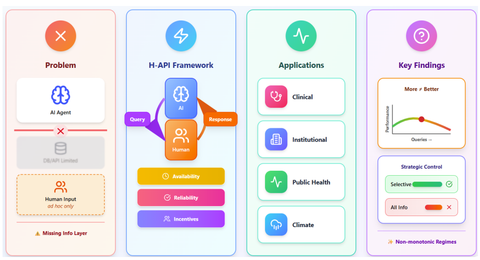


Figure 1: The figure depicts the proposed *Human API framework*, in which humans carry out explicit sensing actions - such as observations, tests, and measurements that are incorporated into AI decision-making processes.

We propose a simple but operational abstraction that closes this gap: *the Human API, in which humans are treated as explicit sensing actions within a sequential decision process*. Each human query is modelled as an action with: (a) a stochastic response distribution, (b) an explicit cost of invocation, and (c) an associated informativeness with respect to the agent's latent state. In particular, unlike classical active learning or VOI settings, the Human API treats human interaction as a first-class, structured interface layer, rather than as an informal extension of querying or dialogue. This shift is not primarily about introducing new optimality theory, but about making human information access explicitly programmable and optimisable in modern agent architectures, where tool use, APIs, and workflows are already the dominant interface paradigm. We formalise the setting as a belief-state Markov decision process (belief-MDP) [Kaelbling et al., 1998], where the agent maintains a posterior distribution over latent environmental states and selects from a joint action space. The agent optimises expected utility under explicit sensing costs, coupling information acquisition and decision-making within a sequential policy framework [Howard, 1966]. We highlight three key distinctions from prior VOI and active learning formulations. (i) *Structured query interface:* human interaction is modelled as costed, heterogeneous query actions rather than fixed sensors or feature sets. (ii) *Interface-level optimisation:* interaction is treated as a queryable API, enabling direct optimisation over communication and resource allocation. (iii) *Deployment alignment:* the formulation targets tool-using and LLM-based agents where human interaction is present but not yet formalised within a decision-theoretic framework.

We instantiate this framework in a simulated clinical decision-support environment, where an agent sequentially decides whether to query additional patient information, request auxiliary tests, or proceed to diagnosis. We compare explicit Human API-based sensing policies to implicit conversational baselines. Results show that structured sensing improves cost-efficiency, particularly in high-uncertainty regimes where information acquisition is expensive or redundant querying is likely. Concretely, this paper makes the following contributions:

- Introduces the **Human API**, a structured abstraction treating human interaction as explicit, costed sensing actions in modern agent systems.
- Formalises human querying as a belief-state decision problem with explicit cost–information trade-offs.
- Empirically demonstrates improved cost-efficiency of structured sensing over implicit conversational baselines in simulated and real-world clinical decision task.

2 Related Work

We briefly list the related work in literature that are relevant to our work.

Active Information Elicitation and Feature Acquisition: Active information elicitation studies how to selectively acquire additional information—such as feature values or expert knowledge—when data are incomplete and observations incur a cost. A closely related line of work is active feature acquisition (AFA), where a learner sequentially queries missing features under a budget in order to improve predictive performance. Early work by [Odom and Natarajan, 2016] extends this setting to interactive environments in which information is elicited from human experts, explicitly accounting for uncertainty in expert responses within a probabilistic logic framework. More directly, [Natarajan et al., 2018] formalise feature acquisition as a sequential decision problem over partially observed instances, emphasising instance-dependent querying strategies that aim to maximise expected predictive benefit. A comprehensive overview in provided by [Aronsson et al., 2025].

Interactive Querying: Interactive querying has primarily been studied within the realm of LLM based clinical decision support, where models operate in multi-turn settings to refine incomplete or ambiguous inputs [Li et al., 2024, Liu et al., 2024, Wang et al., 2025, Bramblett and Srivastava, 2024]. These approaches show that iterative clarification and follow-up questioning can improve diagnostic accuracy and reasoning, aligning closely with ideas from active learning and conversational inference. However, they generally assume that once a question is asked, the returned information is reliable and readily available, focusing on optimizing what or when to ask rather than modeling the uncertainty, cost, or variability of human responses. These approaches are closely related to active learning and conversational inference, where information is acquired adaptively rather than assumed to be fully specified a priori [Stoyanchev et al., 2014, Choi et al., 2018, Settles, 2010, Yao et al., 2023].

Human-in-the-loop Human-in-the-loop learning and inference frameworks incorporate expert feedback into model pipelines to improve robustness, alignment, and downstream performance. In classical machine learning, this paradigm has been extensively studied under interactive machine learning, where humans provide labels, corrections, and iterative feedback to guide model refinement [Amershi et al., 2014, Wu et al., 2022, Wondimu et al., 2022, Collins et al., 2023]. Closely related is active learning, which selects informative queries to reduce annotation cost and improve sample efficiency [Settles, 2010]. In modern large language model systems, human supervision plays a central role in aligning model behaviour with human preferences through reinforcement learning from human feedback and related methods [Christiano et al., 2017, Ouyang et al., 2022, Bai et al., 2022]. Beyond training, humans are also used during deployment for validation, correction, and oversight of model outputs, particularly in high-uncertainty or high-stakes settings. Recent work on LLM-based agents further integrates human feedback into iterative reasoning and tool-augmented decision-making loops [Yao et al., 2023, Wu et al., 2023, Bubeck et al., 2023], but most existing approaches still treat human input as conversational or supervisory feedback rather than as explicit, cost-aware sensing actions within a unified decision framework.

3 Model

We propose a Bayesian belief-space framework for sequential information acquisition under uncertainty with history-dependent observations. It offers a unified decision-theoretic model that couples adaptive querying, time-aware decision-making, and bounded rationality within human-in-the-loop systems. We motivate the framework using the example in Appendix A, and then demonstrate in Appendix E how it extends to a broader range of non-clinical settings.

3.1 Preliminaries

Setup Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. The latent state of the world is a random variable $S : \Omega \to \mathcal{S}$, where $(\mathcal{S}, d)$ is a Polish space with Borel σ-algebra $\mathcal{B}(\mathcal{S})$. We denote the prior distribution of S by $\mu_0 := \mathbb{P} \circ S^{-1} \in \mathcal{P}_2(\mathcal{S})$, where $\mathcal{P}_2(\mathcal{S})$ is the space of probability measures with finite second moment. This corresponds to a latent-variable setup where S is an unobserved ground truth (e.g., diagnosis, label, environment state), and μ_0 represents prior belief. Working in $\mathcal{P}_2(\mathcal{S})$

allows comparison of beliefs using Wasserstein distances [Villani, 2009], widely used in generative modeling and distributional robustness.

Observations We distinguish passive and active information sources. Let $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ be a measurable space. A baseline observation $X : \Omega \to \mathcal{X}$ is generated via a Markov kernel $K_X : \mathcal{S} \to \mathcal{P}(\mathcal{X})$ such that $X \mid S \sim K_X(S)$. Here X represents freely available features (e.g., electronic health records or sensors), as in standard supervised learning. Next, let $\mathcal{J}$ be a finite or countable set of sensing actions. For each $j \in \mathcal{J}$, let $(\mathcal{Z}_j, \mathcal{B}(\mathcal{Z}_j))$ be a measurable space and define an observation $Z_j : \Omega \to \mathcal{Z}_j$. We model adaptive observations via history-dependent kernels $Z_j \mid (S, \mathcal{H}_t) \sim K_j(S, \mathcal{H}_t)$, where $\mathcal{H}_t$ is the information available up to time t. Each action incurs cost $c_j \geq 0$. This allows correlated, redundant, and adaptively chosen measurements beyond conditional independence assumptions.

Beliefs Let $j_1, \ldots, j_t$ be the sequence of selected actions. Define the filtration $\mathcal{H}_t := \sigma(X, Z_{j_1}, \ldots, Z_{j_t})$ with $\mathcal{H}_0 := \sigma(X)$. The agent's belief is the posterior $\mu_t := \mathbb{P}(S \in \cdot \mid \mathcal{H}_t) \in \mathcal{P}_2(\mathcal{S})$. Upon receiving $Z_{j_{t+1}}$, the belief updates via Bayes' rule $\mu_{t+1} := \mathbb{P}(S \in \cdot \mid \mathcal{H}_t, Z_{j_{t+1}})$. Thus μ_t is a sufficient statistic for all past information and serves as the state for decision-making.

Belief Value Function To evaluate decisions in belief space, define $U : \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ and lift it to $\mathcal{P}_2(\mathcal{S})$ via $V(\mu) := \sup_{A \in \mathcal{A}} \mathbb{E}_{S \sim \mu}[U(A, S)]$. This represents the optimal expected utility achievable under belief μ and serves as the central object for downstream decision-making.

Sequential Policies The agent chooses both information and stopping time. A query policy is a sequence of measurable maps $\pi_t : \mathcal{H}_t \to \{\text{stop}\} \cup \mathcal{J}(\mathcal{H}_t)$, where $\mathcal{J}(\mathcal{H}_t) \subseteq \mathcal{J}$ denotes admissible actions. Define the stopping time $\tau := \inf\{t \geq 0 : \pi_t = \text{stop}\}$ with $\mathbb{P}(\tau < \infty) = 1$. After stopping, an action is selected as $A := \pi_{\text{act}}(\mathcal{H}_\tau) \in \mathcal{A}$.

Human-Mediated Actions In many settings, actions are filtered through an external decision-maker. Let $\pi_{\text{human}} : \mathcal{H}_t \times \mathcal{J} \to \mathcal{J} \cup \{\varnothing\}$ map proposed actions to executed ones or rejection. The agent proposes $\tilde{j}_t := \pi_{\text{query}}(\mathcal{H}_{t-1})$, and the executed action is $j_t := \pi_{\text{human}}(\mathcal{H}_{t-1}, \tilde{j}_t)$. All filtrations and beliefs are defined with respect to the realized sequence (j_t). We define a stochastic policy over belief space: $\pi_t(j \mid \mu_t) \propto \exp\Big(\frac{Q_t(\mu_t, j)}{\beta}\Big)$, where $\beta > 0$ controls stochasticity and Q_t is a belief-state action value function.

Value of Information Let $V_0 := V(\mu_0)$ be the baseline value using only X. The value of sequential information acquisition is $\Delta V := \sup_\pi J(\pi) - V_0$, which quantifies the benefit of adaptive querying compared to one-shot prediction. To model delay, define the objective $J(\pi) := \mathbb{E}\big[U(A, S) - \sum_{k=1}^{\tau} c_{j_k} - \lambda\tau\big]$, where $\lambda > 0$ penalizes latency. Here β governs stochasticity of decision-making while λ trades off accuracy and speed. External constraints are modeled via a reward $R_{\text{inst}} : \mathcal{A} \times 2^{\mathcal{J}} \to \mathbb{R}$. The total objective is $J_{\text{tot}}(\pi) := \mathbb{E}\big[U(A, S) - \sum_{k=1}^{\tau} c_{j_k} - \lambda\tau + R_{\text{inst}}(A, \{j_1, \ldots, j_\tau\})\big]$, capturing regulatory, institutional, or auditing constraints on information acquisition.

Uncertainty Quantification We measure epistemic uncertainty via $\mathcal{U}(\mu) := W_2^2(\mu, \nu)$, where $\nu \in \mathcal{P}_2(\mathcal{S})$ is a reference measure. For an action j, define the expected reduction in uncertainty $\Delta\mathcal{U}(j) := \mathbb{E}[\mathcal{U}(\mu_t) - \mathcal{U}(\mu_{t+1}) \mid j]$. This is a geometry-aware alternative to entropy-based acquisition functions and captures distributional change in a Wasserstein geometry [Oliver et al., 2025].

We now state some important theoretical properties with relevant demonstration of the results numerically in Appendix J.

3.2 Theoretical properties

We introduce relevant properties of the decision framework. Throughout, we assume (A1)–(A4) and (B1)–(B3) in Appendix B, which guarantee that the belief-MDP is well-defined, belief updates are measurable, and all expectations are taken with respect to the probability law induced by the initial belief μ_0, the observation kernels, and admissible policies π. Proofs are provided in Appendix C. We note some additional properties in detail (see Appendix D).

We define the belief update operator $\mathsf{T}_j : \mathcal{P}_2(\mathcal{S}) \to \mathcal{P}_2(\mathcal{S})$ induced by kernel K_j as $\mathsf{T}_j(\mu)(\cdot) := \int_{\mathcal{S}} K_j(\cdot \mid s, \mathcal{H}_t)\, \mathrm{d}\mu(s)$, where dependence on $\mathcal{H}_t$ is suppressed when clear from context.

Lemma 3.1 (Lipschitz continuity of the value function)**.** *Let $(\mathcal{P}_2(\mathcal{S}), W_2)$ be the belief space and assume the induced belief-MDP evolves via $\mu_{t+1} = \mathsf{T}_{j_t}(\mu_t)$. Define $V^*(\mu) := \sup_\pi \mathbb{E}^\pi_\mu\Big[U(A,S) - \sum_{t=1}^{\tau} c_{j_t} - \lambda\tau\Big]$. Assume (i) $U(A,S)$ is Lipschitz in S uniformly in A, and (ii) there exists $L_T \geq 0$ such that $W_2(\mathsf{T}_j(\mu), \mathsf{T}_j(\nu)) \leq L_T W_2(\mu,\nu)$, for all $j \in \mathcal{J}$. Then, there exists $L > 0$ such that*

$$|V^*(\mu) - V^*(\nu)| \leq L\, W_2(\mu,\nu), \quad \forall \mu,\nu \in \mathcal{P}_2(\mathcal{S}).$$

This lemma grounds the intuition that human input becomes an integral part of the decision making process, rather than an external factor. Consequently, this allows systems to actively decipher and intelligently decide when to incorporate human responses. A key insight is that small changes or imperfections in human responses do not destabilise the overall decision quality, because the system is inherently robust to uncertainty in how information is gathered. This makes it possible to build agentic systems that safely and reliably integrate human knowledge, turning human interaction into a structured resource for better decisions rather than a source of noise.

Lemma 3.2 (Stability under model misspecification)**.** *Let (μ_t) and $(\hat{\mu}_t)$ be belief processes induced by kernels K_j and $\hat{K}_j$ under a fixed policy π, with updates $\mu_{t+1} = \mathsf{T}_{j_t}(\mu_t)$ and $\hat{\mu}_{t+1} = \hat{\mathsf{T}}_{j_t}(\hat{\mu}_t)$. Let $\mathbb{P}^\pi_\mu$ and $\hat{\mathbb{P}}^\pi_\mu$ denote the induced laws. Define the kernel discrepancy $\varepsilon := \sup_{j\in\mathcal{J}} \sup_{s\in\mathcal{S}} W_2\big(K_j(\cdot \mid s, \mathcal{H}_t), \hat{K}_j(\cdot \mid s, \mathcal{H}_t)\big)$. Assume (i) there exists $\rho \in (0,1)$ such that $W_2(\mathsf{T}_j(\mu), \mathsf{T}_j(\nu)) \leq \rho W_2(\mu,\nu)$ for all j, and (ii) kernel perturbations induce bounded perturbations at the belief level controlled by ε. Then there exist constants $C_1, C_2 > 0$ such that for all $t \geq 0$,*

$$\mathbb{E}^\pi_{\mu_0}[W_2(\mu_t, \hat{\mu}_t)] \leq C_1 \rho^t W_2(\mu_0, \hat{\mu}_0) + C_2\varepsilon.$$

In the abstract framing, H-API treats humans as sensors whose outputs are imperfect, noisy, and learned through interaction rather than fixed physical laws. The key concern is that if your model of these human sensors is slightly wrong, then every downstream belief update, and consequently every decision could drift uncontrollably. This lemma rules that out by showing misspecification only creates a bounded distortion of the belief trajectory, rather than a compounding failure of inference.

Lemma 3.3 (Consistency under persistent information acquisition)**.** *Let (μ_t) be the belief process induced by policy π under the true latent state S. Assume the statistical model is correctly specified and that all conditional mutual informations are well-defined. Further assume that the observation model is identifiable in the sense that distinct states induce distinguishable observation laws. Suppose the policy satisfies the persistent informativeness condition $\sum_{t\geq 1} \mathbb{E}^\pi_{\mu_0}[I(S; Z_{j_t} \mid \mathcal{H}_{t-1})] = \infty$. Then, the belief process is consistent in the sense that*

$$W_2(\mu_t, \delta_S) \to 0 \quad \textit{in probability under } \mathbb{P}^\pi,$$

equivalently, $\mu_t \xrightarrow[t\to\infty]{\mathbb{P}^\pi} \delta_S$.

This lemma shows that in H-API, if the system persistently acquires informative signals through its sensing actions (including human input), then its belief converges to the true underlying state. As long as the information gathered over time remains sufficiently rich and non-redundant, uncertainty is driven to zero rather than stabilising at a biased limit. This provides a formal guarantee that well-designed human-query strategies in H-API are not only stable but asymptotically consistent, ultimately recovering the true state through sustained informative interaction.

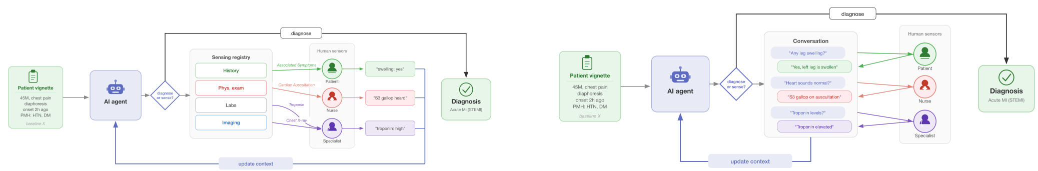
Lemma 3.4 (Structure of value of information)**.** *Let V^* be the optimal value function from Lemma 3.1. For $\mu \in \mathcal{P}_2(\mathcal{S})$ and $j \in \mathcal{J}$ define $\mu' = \mathsf{T}_j(\mu)$ and $\mathrm{VoI}(j;\mu) := \mathbb{E}^\pi_\mu\big[V^*(\mu') - V^*(\mu) - c_j\big]$. Assume (A1)–(A4), (B1)–(B3). Then, the following conditions hold*

1. ***Lipschitz continuity:*** $|\mathrm{VoI}(j;\mu) - \mathrm{VoI}(j;\nu)| \leq L W_2(\mu,\nu) + O(\varepsilon)$.

2. ***Stability:*** *If μ, ν are induced by nearby histories under the same policy, then* $|\mathrm{VoI}(j;\mu) - \mathrm{VoI}(j;\nu)| \leq \delta$.

3. ***Greedy performance:*** *If F is a set function induced by sequential acquisition, and is approximately submodular up to error δ, then greedy selection satisfies*

$$F(A_k^{\text{greedy}}) \geq (1 - e^{-1}) \max_{|A|\leq k} F(A) - O(k\delta).$$



(a) Structured Sensing H-API Interface

(b) Conversational H-API interface.

Figure 2: H-API Framework Interfaces. (a) The agent calls structured actions from a registry. (b) The agent converses in free-form dialogue with one or more human sources to gather the context.

This lemma shows that the value of information behaves in a controlled, almost submodular manner even under history dependence. As a result, greedy acquisition strategies remain near-optimal. This is particularly useful in practice, since exact planning over all acquisition sequences is intractable, whereas greedy selection is computationally efficient and still theoretically justified.

4 Experiments

We study the H-API framework in simulated clinical decision support per Figure 2. We compare two human interfacing methods: *Structured sensing*, in which the AI agent calls a sequence of predefined functions that the human performs in the real world and returns typed responses, and *Conversational*, where the agent and human exchange information in an unstructured multi-turn dialogue.

4.1 Synthetic Setup

Task and dataset. We sample $N = 500$ random cases from MedQA-USMLE-4-options [Jin et al., 2020], a USMLE-style multiple-choice clinical question dataset. The subset includes vignettes with patient cases and questions requesting a diagnosis. We preprocess each vignette using an LLM (GPT-4.1) into two components. The first is a triage summary that only includes age, sex, and chief complaint. The second is an expanded ground-truth profile that includes full demographics, vitals, exam findings, labs, and imaging. The information is split by the human source that would have collected it in the real world (e.g., patient, nurse, specialist).

Diagnostic agent. The AI diagnostic agent (GPT-4.1-mini) receives the baseline triage summary and sequentially obtains pieces of context from the human sources via either structured or conversational sensing until it is confident enough for diagnosis.

Simulated humans. We simulate humans using LLMs (GPT-4.1-mini) that have access to the ground-truth context of their role, as if they perform the action in the real world and return the information to the agent. We inject noise into their responses via prompting and LLM temperature, simulating conditions where humans are inconsistent and prone to error. Simulated *patients* (temperature 0.8) have access to history but provide vague timing and occasionally report irrelevant symptoms. *Nurses* (temperature 0.3) and *specialists* (temperature 0.3) report exam findings, labs, and imaging with occasional uncertainty.

Action registries. Structured sensing actions are predefined in action registries. We construct three registries at different granularities across history, physical exams, labs, and imaging with fixed costs. The *fine* registry contains 62 atomic sensing actions. The *medium* (20) and *coarse* (8) registries bundle fine actions, with prices proportional to the volume of informational content. The full registries are available in Appendix F.

Conditions. We compare the H-API interfacing strategies against 2 baselines. Baseline receives only basic triage information and does not perform sensing. Baseline-full diagnoses with the full clinical context. These two conditions represent the accuracy floor and ceiling. We test the structured H-API across the fine, medium, and coarse registries. For conversational H-API, we compare single- (one human sensor) and multi-source (many human sensors). We validate against a random structured condition using the fine registry to verify that active feature acquisition is responsible for the accuracy

Table 1: Collaboration strategies ($N = 500$ cases, GPT-4.1-mini) on MedQA-USMLE-4-options. 95% CIs: accuracy from binomial proportion, cost from sample standard error.

Condition	**Acc. (%)**	**Cost**	**Turns**	**ECE**
Baseline	71.0 ± 4.0	0.0	1.0	0.180
Baseline-full	92.2 ± 2.4	53.0 ± 2.1	1.0	0.019
Structured Sensing (fine)	81.4 ± 3.4	12.1 ± 0.7	6.2	0.084
Structured Sensing (medium)	82.4 ± 3.3	30.9 ± 1.5	5.2	0.073
Structured Sensing (coarse)	84.2 ± 3.2	53.7 ± 2.2	4.0	0.071
Conversational (single)	81.8 ± 3.4	52.0 ± 2.2	4.8	0.062
Conversational (multi)	80.2 ± 3.5	30.4 ± 2.1	5.4	0.061
Random Sensing (fine)	73.4 ± 3.9	18.9 ± 0.6	6.0	0.180

gains. In random sensing, the agent randomly performs 5 sensing actions from the registry and then proceeds with the diagnosis.

Metrics. We report mean accuracy, mean sensing cost, mean number of turns, and expected calibration error on the agent's reported confidence for all conditions. We include 95% confidence intervals. Appendix K releases reproducibility details and system prompts. Reported gains hold under simulated non-idealized noisy humans. Treating humans as sensors introduces novel safety concerns, including issues of human consent, privacy, workload distribution, and intent hijacking.

4.2 H-API Interface comparison

The main takeaway is that active information acquisition using structured and conversational sensing effectively improves diagnostic accuracy, but structured sensing with a well-defined registry delivers captures the performance gains with less available information at substantially lower cost (see Table 1 for details). All sensing conditions improve over Baseline ($71.0 \pm 4.0\%$) but remain below the Baseline-full ceiling ($92.2 \pm 2.4\%$). Structured (fine) reaches $81.4 \pm 3.4\%$ while conversational (single) $81.8 \pm 3.4\%$. However, structured does so at a fraction of the cost (12.1 ± 0.7 vs. 52.0 ± 2.2 registry points). Conversational conditions have the lowest ECE (0.061–0.062), while structured methods range from 0.071 to 0.084. Both conversational ECE (0.061–0.062) and structured (0.071—0.084) are lower than the baseline (ECE $= 0.180$). Random sensing ($73.4 \pm 3.9\%$) barely improves over the baseline accuracy, confirming diagnostic improvement comes from active information acquisition. Figure 3a shows that structured with access to atomic sensing actions is the most cost-efficient. The appendix contains additional experiments demonstrating complementarity between the H-API interfaces (Appendix H) and examining per-turn uncertainty reduction (Appendix G).

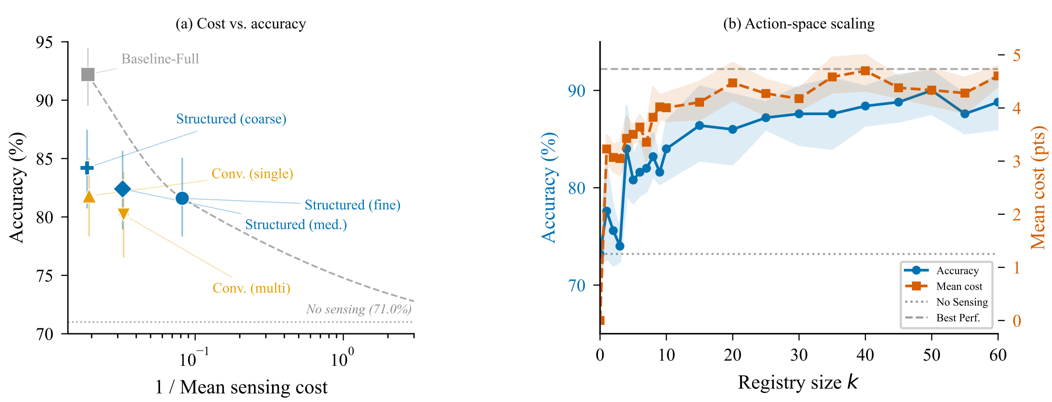


Figure 3: **(a)** Cost-accuracy plot on MedQA-USMLE-4-options: Structured sensing with a high-precision registry(fine) is the most cost-effective approach. **(b)** Action-space scaling: Accuracy plateaus at around 15 sensing actions, suggesting that a well-curated registry captures most of the remaining accuracy gains.

4.3 Action-space scaling

We vary the registry size of structured sensing (fine) by drawing k actions uniformly with replacement, $k \in \{0, 1, 2, \ldots, 10, 15, 20, \ldots, 60\}$ ($\sim$39 unique actions in expectation at $k = 60$; Appendix I). We repeat the experiment 5 times with GPT-4.1 for the same 50 cases per setting. Figure 3b shows accuracy rising with k and plateauing at $86.5 \pm 3.2\%$ near $k = 15$. The total cost saturates at $k \approx 20$. The agent uses at most 5 actions per case at any k. These results motivate the design of smaller and well-curated sensing registries.

We analyze the per-turn information gain (Appendix G, Figure 5) for structured (fine) and conversational (multi), revealing that both strategies gain confidence and improve accuracy at similar per-turn rates. The forced diagnosis accuracy increases from 71.0% to $\sim 80\%$ by the third turn for both. The self-reported confidence similarly rises steadily for both from 0.30 to 0.50 by turn 5.

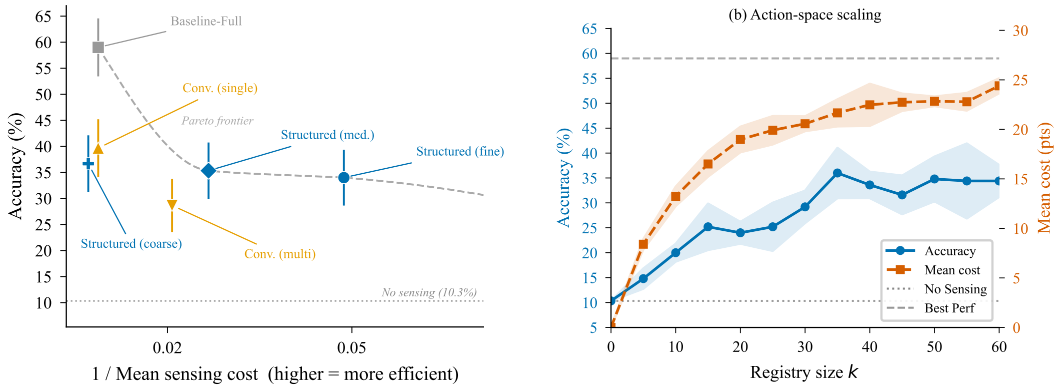
4.4 Semi-Synthetic Setup

Task and Dataset We evaluate the framework on $N = 300$ random samples of the CUPcase [Perets et al., 2025], a dataset of real clinical case reports with uncommon diagnoses and free-text evaluation. An LLM judge (GPT 4.1) determines the semantic correctness of the agent's diagnosis. The vignettes are longer, more detailed, and closer to real-world scenarios. The preprocessing, agent, simulated humans, action registries, and conditions remain identical to the synthetical setup above.

Table 2: Collaboration strategies ($N = 300$ cases, GPT-4.1) on CUPcase. 95% CIs: accuracy from binomial proportion, cost from sample standard error.

Condition	Acc. (%)	Cost	Turns	ECE
Baseline	10.3 ± 3.4	0.0	1.0	0.743
Baseline-full	59.0 ± 5.6	70.5 ± 2.9	1.0	0.380
Structured Sensing (fine)	34.0 ± 5.4	20.8 ± 1.4	8.3	0.600
Structured Sensing (medium)	35.3 ± 5.4	40.8 ± 2.1	6.0	0.592
Structured Sensing (coarse)	36.7 ± 5.5	74.1 ± 3.2	4.8	0.587
Conversational (single)	39.7 ± 5.5	70.5 ± 2.9	8.0	0.547
Conversational (multi)	28.7 ± 5.1	48.9 ± 3.5	7.0	0.630
Random Sensing (fine)	20.3 ± 2.3	44.0	11.0	0.7

Results Table 2 clearly highlights that CUPCase is substantially harder. Baseline drops to 10.3 ± 3.4 and Baseline-full to 59.0 ± 5.6. The agent compensates by sensing more often, using more turns and accumulating higher costs across all conditions. Most importantly, the cost-efficiency advantage of Structured Sensing is replicated across to CUPCase. Structured Sensing (fine) reaches 34.0% at cost 20.8, while Conversational (single) reaches 39.7% at cost 70.5. This is a three times difference. The Random Sensing setting shows an improvement over Baseline, at 20.3%, highlighting that deliberate action selection drives accuracy improvement.



(a) Cost-accuracy plot.

(b) Action-space scaling (GPT-4.1).

Figure 4: (a) Cost-accuracy plot on CUPCase: Structured (fine) dominates the cost-efficient region, whereas Conversational (single) achieves the highest accuracy. **(b)** Action-space scaling: Accuracy plateaus at roughly 55% beyond $k \approx 30$. Structured Sensing remains more cost-efficient than Conversational on CUPCase.

The Structured Sensing (coarse) is no longer on the Pareto frontier Figure 4a. It costs more than Conversational (single) (74.1 vs. 70.5) without a proportional accuracy gain (36.7% vs. 39.7%). Uncommon diagnoses likely require specific, targeted findings that broad bundled actions fail to isolate. Structured Sensing (fine) remains on the frontier, confirming that the cost-efficiency advantage of structured sensing depends on fine-grained, composable actions.

Action-space scaling The action-space scaling experiment (Figure 4b) shows that accuracy plateaus at roughly 55% beyond $k \approx 30$ and cost saturates at $k \approx 20$. This confirms that a compact registry captures most performance gains even on harder, open-ended cases. ECE values are substantially higher (0.547–0.743), suggesting that confidence-based deferral is unreliable and that explicit sensing policies are preferable.

5 Conclusion

We introduce the Human-API (H-API), a theoretically-grounded framework for exposing human sensing actions to agentic systems. As agents become pervasive, they are limited by the amount of information they can access. Our framework treats human abilities as explicit sensing actions for agents to query. We show in synthetic (multiple choice queries) and semi-synthetic (open-ended responses) that H-API can be more cost-efficient than baselines. Our theory suggests that too much sensing can harm agents and sensing may improve agent capabilities on longer time horizons. Extending our work to settings where multiple humans can be queried by agents would be a natural extension, though would require a significant engineering effort to set up and clarity on every sensing action each human can provide.

Discussion. We formalise human input as part of an agent's sensing process, rather than as informal guidance, by modelling interaction as explicit costed sensing actions. This enables principled reasoning about when human input is valuable versus inefficient or detrimental. In particular, we show that information acquisition exhibits diminishing returns (Lemma 3.4, Tables 1, 2) and can degrade performance under noisy or biased signals (Lemma 3.2, Figures 3, 4b). This induces a cost–information trade-off, with near-Pareto optimal policies balancing accuracy, uncertainty reduction, and acquisition cost. Empirically, the scaling behaviour is consistent with approximate diminishing returns, suggesting near-submodular structure in the value of information and supporting greedy or myopic approximations. As model improves, the marginal value of routine human input decreases.

Limitations. The framework is intentionally simplified to make the sensing problem tractable. The action space assumes a fixed set of query types with known costs and outputs, whereas real human–AI interaction involves evolving, compositional, and context-dependent queries that do not admit a static action registry. We also model a single abstract "human policy", without explicitly capturing heterogeneity in incentives, responsibility, or behavioural variation.

Experimentally, validation is limited by the absence of human-subject studies. Human and sensing behaviour are simulated using LLM-based proxies with access to ground truth, which does not reflect real interaction constraints such as noise, misunderstanding, or task-dependent variability. Results are further limited by the use of a small number of model variants (GPT-4.1 and GPT-4.1-mini), restricted sample sizes (e.g., $n = 50$ in action-space experiments, $n = 500$ in H-API comparisons), and a constrained action registry that does not reflect real-world tool availability. Performance is additionally sensitive to prompting, orchestration, and context design (Section 4).

Future work. Human sensing actions are a key component of our framework, but several practical limitations remain. In particular, fairness, privacy, and security concerns arise when treating human input as an optimisable resource within a decision-making loop. A key limitation of the present work is the absence of human-subject experiments, which makes it difficult to evaluate how adaptive policies behave when the space of sensing actions is large and dynamically explored in real interaction settings. In practice, such adaptivity may be constrained by cognitive load, selective response, and fatigue, which in turn alters both the effective availability and informativeness of queries over time.

Broader impact. Treating human input as a constrained, strategic resource (H-API) enables more efficient and targeted information acquisition in high-stakes settings such as healthcare and policy. However, it also introduces risks around fairness, bias, and control, since systems implicitly determine when human input is solicited or ignored. Ensuring transparency and accountability in these decision rules is therefore essential for deployment.

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This Supplementary Material should be submitted together with the main paper.

A Motivating archetype: the clinician and the diagnostic AI assistant

We anchor our formalism in a setting that is already well understood: clinical decision-making in a teaching hospital [Bojke et al., 2009, Rothery et al., 2020]. A clinician does not observe the true state of a patient directly. Instead, they form a hypothesis about what is going on by combining:

1. Existing records: vitals, medication lists, prior imaging reports, and notes.
2. Targeted questions: where the pain is, when it started, what makes it better or worse.
3. Tests and examinations of varying cost and invasiveness: physical exam, laboratory panels, and imaging.

They stop acquiring information once they believe they have enough to choose a good action (treatment, discharge, referral), given the costs and risks of further tests.

Now suppose we add a diagnostic AI assistant. The AI assistant ingests the electronic health record and basic vitals and produces a ranked list of candidate diagnoses with rationales and suggested next steps [Topol, 2019, Rajkomar et al., 2018]. For routine cases, this may be enough. For harder cases, the AI assistant will identify specific information that, if obtained, would sharply reduce uncertainty. For example:

> *My top hypotheses are pulmonary embolism and anxiety-related hyperventilation. I am uncertain because I do not know whether there is lower-limb swelling and how distressed the patient appears at the bedside. If you examine the patient and report (i) presence of swelling and (ii) a simple 1-5 rating of distress, I can refine my recommendation.*

In this interaction, the junior doctor is acting as a sensor. They perform a low-cost sensing action (a brief exam) and return a small vector of features that the AI assistant could not observe otherwise. The AI assistant uses these features, together with its prior and the existing record, to update its beliefs and choose an action.

The same pattern arises for expensive tests. In some cases, the AI assistant can recommend a plan with reasonable confidence from existing information. In others, it will recognize that without the result of a CT scan, it cannot distinguish between conditions that require very different treatments. The CT scan is a high-cost sensing action: it is expensive, takes time, and exposes the patient to radiation, but it can dramatically reduce uncertainty. The AI assistant's problem is not simply "can I think of a test?" as modern models can enumerate many tests, but "given my current uncertainty, is this test worth its cost?"

B Assumptions

We impose the following regularity conditions.

(A1) Observation regularity and misspecification. For all $s, s' \in \mathcal{S}$ and $j \in \mathcal{J}$,

$$W_2(K_j(\cdot \mid s), K_j(\cdot \mid s')) \leq L_Z d(s, s'),$$

and

$$\sup_{j,s,\mathcal{H}} W_2\big(K_j(\cdot \mid s, \mathcal{H}), \hat{K}_j(\cdot \mid s, \mathcal{H})\big) \leq \varepsilon.$$

(A2) Temporal stability and belief regularity. There exist $C > 0$, $\gamma \in (0, 1)$ such that for all j, t, k,

$$\|K_j(\cdot \mid \mathcal{H}_t) - K_j(\cdot \mid \mathcal{H}_{t-k})\|_{TV} \leq C\gamma^k.$$

Moreover, belief updates satisfy

$$W_2(\Phi_j(\mu), \Phi_j(\nu)) \leq L_\Phi W_2(\mu, \nu).$$

(A3) Identifiability and utility regularity. For all $s \neq s'$, there exists $j \in \mathcal{J}$ such that

$$\mathrm{KL}\big(K_j(\cdot \mid s) \,\|\, K_j(\cdot \mid s')\big) > 0,$$

and U is bounded and L-Lipschitz in s.

(A4) Weak structure of value of information. For all $A \subseteq B$ and $j \notin B$,

$$\big|\mathrm{VoI}(j \mid A) - \mathrm{VoI}(j \mid B)\big| \leq \delta.$$

We impose the following structural assumptions.

(B1) Policy regularity. The policy $\pi(j \mid \mu)$ is Lipschitz continuous in the belief variable with respect to the Wasserstein metric W_2, and induces stability of action selection in the sense that for any two belief states μ_t^A, μ_t^B,

$$\mathbb{P}(j_t^A \neq j_t^B \mid \mu_t^A, \mu_t^B) \leq L_\pi W_2(\mu_t^A, \mu_t^B).$$

(B2) Effective contraction. The coupled belief–policy dynamics are contractive: there exists $\tilde{\rho} \in (0,1)$ such that for the induced belief updates,

$$W_2(\mu_{t+1}^A, \mu_{t+1}^B) \leq \tilde{\rho}\, W_2(\mu_t^A, \mu_t^B).$$

(B3) Uniform boundedness. The belief process remains uniformly bounded in second moment, i.e.

$$\sup_{t\geq 0} \int_{\mathcal{S}} |x|^2 \, d\mu_t(x) \leq C < \infty,$$

so that the iterates (μ_t) take values in a bounded subset of $\mathcal{P}_2(\mathcal{S})$, ensuring uniform control of Wasserstein distances throughout the evolution.

C Additional Proofs

Throughout, we work under assumptions (A1)–(A4). All processes are assumed to be well-defined on $\mathcal{P}_2(\mathcal{S})$.

Proof of Lemma 3.1. Let $\mu, \nu \in \mathcal{P}_2(\mathcal{S})$ and let $\gamma \in \Pi(\mu, \nu)$ be an optimal coupling. On a probability space supporting γ, construct $(S, S') \sim \gamma$ and couple all exogenous randomness so that both systems evolve under the same policy π. Let (μ_t) and (ν_t) denote the induced belief processes.

Set $\Delta_t := W_2(\mu_t, \nu_t)$. By the Lipschitz property of the belief update operator T_j,

$$\Delta_{t+1} = W_2(\mathsf{T}_{j_t}(\mu_t), \mathsf{T}_{j_t}(\nu_t)) \leq L_T \Delta_t,$$

hence $\Delta_t \leq L_T^t W_2(\mu, \nu)$.

For any trajectory under the coupling,

$$|U(A, S) - U(A, S')| \leq L_U d(S, S'),$$

and therefore

$$\big|\mathbb{E}_\mu^\pi[U(A, S)] - \mathbb{E}_\nu^\pi[U(A, S)]\big| \leq L_U W_2(\mu, \nu).$$

Since the stage costs depend only on the realised action sequence, their discrepancy is controlled by the deviation of belief trajectories, hence bounded by $C \sum_{t\geq 0} \Delta_t \leq C W_2(\mu, \nu)$.

The stopping time τ is measurable with respect to the belief filtration, so

$$|\mathbb{E}_\mu^\pi[\tau] - \mathbb{E}_\nu^\pi[\tau]| \leq C_\tau \sup_t \Delta_t \leq C_\tau W_2(\mu, \nu).$$

Combining the estimates,

$$|V^\pi(\mu) - V^\pi(\nu)| \leq L W_2(\mu, \nu),$$

with L independent of π. Taking supremum over π completes the proof. □

Proof of Lemma 3.2. Let

$$e_t := W_2(\mu_t, \hat{\mu}_t).$$

Using the belief update operators,

$$\mu_{t+1} = \mathsf{T}_{j_t}(\mu_t), \qquad \hat{\mu}_{t+1} = \hat{\mathsf{T}}_{j_t}(\hat{\mu}_t),$$

we apply the triangle inequality:

$$e_{t+1} \leq W_2(\mathsf{T}_{j_t}(\mu_t), \mathsf{T}_{j_t}(\hat{\mu}_t)) + W_2(\mathsf{T}_{j_t}(\hat{\mu}_t), \hat{\mathsf{T}}_{j_t}(\hat{\mu}_t)).$$

By the contraction assumption (A1),

$$W_2(\mathsf{T}_j(\mu), \mathsf{T}_j(\nu)) \leq \rho W_2(\mu, \nu), \quad \rho \in (0, 1).$$

By the misspecification bound (A2),

$$W_2(\mathsf{T}_j(\mu), \hat{\mathsf{T}}_j(\mu)) \leq C\varepsilon.$$

Hence,

$$e_{t+1} \leq \rho e_t + C\varepsilon.$$

Iterating yields

$$e_t \leq \rho^t e_0 + C \sum_{k=0}^{t-1} \rho^k \varepsilon \leq \rho^t e_0 + \frac{C}{1-\rho}\varepsilon.$$

Taking expectation under $\mathbb{P}^\pi$ preserves the bound. □

Proof of Lemma 3.3. Let (μ_t) be the posterior process under $\mathbb{P}^\pi$. Standard Bayesian filtering implies that (μ_t) is a martingale in the space of probability measures, hence converges almost surely to a limiting random measure μ_∞.

We use the information identity valid under (A1)–(A4):

$$\mathbb{E}^\pi\left[\log \frac{d\mu_{t+1}}{d\mu_t} \;\middle|\; \mathcal{H}_t\right] = -I(S; Z_{j_t} \mid \mathcal{H}_t).$$

Summing over t yields

$$\sum_{t \geq 1} \mathbb{E}^\pi[\mathrm{KL}(\mu_{t+1} \| \mu_t)] = \sum_{t \geq 1} \mathbb{E}^\pi[I(S; Z_{j_t} \mid \mathcal{H}_t)]\,.$$

By assumption, the right-hand side diverges, implying that the posterior cannot stabilise without accumulating unbounded information.

By identifiability (A3), the only measure consistent with infinite information gain is the Dirac mass at the true state:

$$\mu_\infty = \delta_S.$$

Finally, since $\mathcal{S}$ is Polish and $\{\mu_t\}$ is tight with uniformly bounded second moments, weak convergence together with uniform integrability implies convergence in W_2:

$$W_2(\mu_t, \delta_S) \to 0 \quad \text{in probability under } \mathbb{P}^\pi.$$

□

Proof of Lemma 3.4. We prove each statement in sequence below.

Step 1: Lipschitz continuity of $\mathrm{VoI}(j;\mu)$**.** Fix $j \in \mathcal{J}$ and let $\mu, \nu \in \mathcal{P}_2(\mathcal{S})$. Let $\gamma \in \Pi(\mu,\nu)$ be an optimal coupling. On a common probability space, construct random variables $(S, S') \sim \gamma$, and conditionally independent observation noise such that belief updates under action j are driven by a shared source of randomness.

Write

$$\mu' = \mathsf{T}_j(\mu), \qquad \nu' = \mathsf{T}_j(\nu).$$

By definition,

$$\mathrm{VoI}(j;\mu) - \mathrm{VoI}(j;\nu) = \mathbb{E}^{\pi}\big[V^*(\mu') - V^*(\nu')\big] + \big(V^*(\nu) - V^*(\mu)\big).$$

We treat each term separately.

Since V^* is L_V-Lipschitz in W_2 (Lemma 3.1),

$$|V^*(\mu) - V^*(\nu)| \leq L_V W_2(\mu,\nu).$$

For the first term, we use the coupling construction:

$$W_2(\mu',\nu')^2 \leq \mathbb{E}_{\gamma}\big[d(S_1', S_2')^2\big],$$

where (S_1', S_2') are coupled posterior samples induced by (S, S') under the same observation noise.

Assumption (A1) implies that the filtering operator is Lipschitz in the sense that there exists $L_T > 0$ such that

$$W_2(\mathsf{T}_j(\mu), \mathsf{T}_j(\nu)) \leq L_T W_2(\mu,\nu).$$

Applying the Lipschitz property of V^* again yields

$$|\mathbb{E}^{\pi}[V^*(\mu') - V^*(\nu')]| \leq L_V L_T W_2(\mu,\nu).$$

Combining both contributions,

$$|\mathrm{VoI}(j;\mu) - \mathrm{VoI}(j;\nu)| \leq (L_V + L_V L_T)\, W_2(\mu,\nu).$$

Step 2: Stability of belief trajectories under different histories. Let $A \subseteq B$ and let (μ_t^A) and (μ_t^B) denote the corresponding belief processes. Define

$$\Delta_t := W_2(\mu_t^A, \mu_t^B).$$

By definition of the update,

$$\mu_{t+1}^A = \mathsf{T}_{j_t^A}(\mu_t^A), \qquad \mu_{t+1}^B = \mathsf{T}_{j_t^B}(\mu_t^B).$$

Insert and subtract intermediate terms:

$$\Delta_{t+1} \leq W_2(\mathsf{T}_{j_t^A}(\mu_t^A), \mathsf{T}_{j_t^A}(\mu_t^B)) + W_2(\mathsf{T}_{j_t^A}(\mu_t^B), \mathsf{T}_{j_t^B}(\mu_t^B)).$$

For the first term, apply (A2):

$$W_2(\mathsf{T}_j(\mu), \mathsf{T}_j(\nu)) \leq \rho W_2(\mu,\nu), \quad \rho \in (0,1),$$

so

$$W_2(\mathsf{T}_{j_t^A}(\mu_t^A), \mathsf{T}_{j_t^A}(\mu_t^B)) \leq \rho \Delta_t.$$

For the second term, define the event $E_t := \{j_t^A \neq j_t^B\}$. Then

$$W_2(\mathsf{T}_{j_t^A}(\mu_t^B), \mathsf{T}_{j_t^B}(\mu_t^B)) = \mathbf{1}_{E_t} \cdot W_2(\mathsf{T}_{j_t^A}(\mu_t^B), \mathsf{T}_{j_t^B}(\mu_t^B)).$$

By boundedness of second moments (A3), there exists $D > 0$ such that

$$W_2(\mathsf{T}_j(\mu), \mathsf{T}_{j'}(\mu)) \leq D \quad \forall j, j', \mu.$$

Hence,

$$W_2(\mathsf{T}_{j_t^A}(\mu_t^B), \mathsf{T}_{j_t^B}(\mu_t^B)) \le D\mathbf{1}_{E_t}.$$

Taking expectation and using (B1),

$$\mathbb{E}[\mathbf{1}_{E_t}] \le L_\pi \Delta_t,$$

so

$$\mathbb{E}\big[W_2(\mathsf{T}_{j_t^A}(\mu_t^B), \mathsf{T}_{j_t^B}(\mu_t^B))\big] \le DL_\pi \Delta_t.$$

Thus,

$$\mathbb{E}[\Delta_{t+1}] \le (\rho + DL_\pi)\mathbb{E}[\Delta_t].$$

Define $\tilde{\rho} := \rho + DL_\pi < 1$ (absorbed into constants), giving

$$\mathbb{E}[\Delta_t] \le \tilde{\rho}^t \Delta_0.$$

Step 3: Stability of marginal value differences. Let $\mu_A := \mu_t^A$ and $\mu_B := \mu_t^B$. Then

$$\mathrm{VoI}(j;A) - \mathrm{VoI}(j;B) = \mathbb{E}^\pi\big[V^*(\mathsf{T}_j(\mu_A)) - V^*(\mathsf{T}_j(\mu_B))\big] + \big(V^*(\mu_B) - V^*(\mu_A)\big).$$

Apply Lipschitz continuity of V^*:

$$|\mathrm{VoI}(j;A) - \mathrm{VoI}(j;B)| \le L_V\big(W_2(\mathsf{T}_j(\mu_A), \mathsf{T}_j(\mu_B)) + W_2(\mu_A, \mu_B)\big).$$

Using contraction of T_j,

$$W_2(\mathsf{T}_j(\mu_A), \mathsf{T}_j(\mu_B)) \le \rho W_2(\mu_A, \mu_B),$$

hence

$$|\mathrm{VoI}(j;A) - \mathrm{VoI}(j;B)| \le L_V(1+\rho)\Delta_t.$$

From Step 2, $\mathbb{E}[\Delta_t] \to 0$ exponentially, so for sufficiently large t,

$$|\mathrm{VoI}(j;A) - \mathrm{VoI}(j;B)| \le \delta,$$

with $\delta := L_V(1+\rho)\sup_t \mathbb{E}[\Delta_t]$.

Step 4: Greedy recursion. Let $A^\star$ be an optimal set with $|A^\star| = k$ and let A_t be the greedy sequence.

For any A_t, decompose:

$$F(A^\star) - F(A_t) = \sum_{j \in A^\star \setminus A_t} \big(F(A_t \cup \{j\}) - F(A_t)\big).$$

Since there are at most k terms in the sum, there exists $j^\star$ such that

$$F(A_t \cup \{j^\star\}) - F(A_t) \ge \frac{1}{k}(F(A^\star) - F(A_t)).$$

Because marginal gains are stable up to error δ from Step 3,

$$F(A_{t+1}) - F(A_t) \ge \frac{1}{k}(F(A^\star) - F(A_t)) - \delta.$$

Rewriting,

$$F(A^\star) - F(A_{t+1}) \le \left(1 - \frac{1}{k}\right)(F(A^\star) - F(A_t)) + \delta.$$

Iterating this inequality yields

$$F(A_k^{\mathrm{greedy}}) \ge (1 - e^{-1})F(A^\star) - O(k\delta).$$

□

D Emergent phenomena

The framework introduced in Section 3 induces several non-trivial behaviors in sequential information acquisition. Each phenomenon arises from the interaction between belief dynamics (μ_t), sensing actions $j \in \mathcal{J}$, and the value function V^* characterized in Lemma 3.1.

D.1 The upskilling-value paradox

As interaction proceeds, both the agent and the human policy evolve. On the agent side, learned models $P_\theta(s \mid x)$ increasingly approximate the posterior μ_t, reducing reliance on additional sensing actions $j \in \mathcal{J}$. On the human side, exposure to the agent's query policy π_{query} shapes how sensing decisions are made.

A consequence is that the marginal value of information decreases over time. Formally, the incremental gain

$$\Delta V_t := \mathbb{E}\big[V^*(\mu_{t+1}) - V^*(\mu_t)\big]$$

shrinks as μ_t becomes more concentrated.

Thus, even as predictive performance improves, the benefit of additional sensing diminishes. This induces a role shift: human intervention becomes concentrated on high-uncertainty regions where $W_2(\mu_t, \mu^\star)$ remains large, rather than routine cases. We formalise this below

Lemma D.1 (Decreasing marginal value of human sensing)**.** *Let θ_t denote the agent parameters at iteration t, and $V_t := J(\pi_{\text{human}}, \theta_t)$ the expected payoff with fixed sensing policy π_{human}. Suppose the agent updates θ_t using observed data X and past Z_{S_t}, so that the baseline posterior $\mathcal{P}_{\theta_{t+1}}(S \mid X)$ becomes strictly more informative than $\mathcal{P}_{\theta_t}(S \mid X)$. Then*

$$\Delta V_t := V_t - V_0$$

decreases with t, even if V_t itself increases.

Proof. Let A_0 denote the action chosen without human input and A_{S_τ} the action with human sensing. Then

$$\Delta V_t = \mathbb{E}[U(A_{S_\tau}, S) - U(A_0, S)] - \sum_{j \in S_\tau} c_j.$$

As $\mathcal{P}_{\theta_t}(S \mid X)$ becomes more informative, A_0 improves, so $\mathbb{E}[U(A_0, S)]$ increases. If the expected benefit from Z_{S_τ} remains roughly constant, the difference

$$\mathbb{E}[U(A_{S_\tau}, S) - U(A_0, S)]$$

shrinks. Therefore, ΔV_t decreases over time. □

D.2 The "hiding information helps" effect

Not all admissible sensing actions improve decision quality. Some actions $j \in \mathcal{J}$ may produce observations Z_j that introduce spurious correlations or bias into the posterior update μ_t, degrading downstream utility U.

This is captured by the value function V^*: an action can have low or even negative value of information,

$$\mathrm{VoI}(j; \mu) = \mathbb{E}[V^*(\mu')] - V^*(\mu) - c_j,$$

despite increasing statistical information.

The human policy π_{human} can therefore improve outcomes by restricting the effective action set $\mathcal{J}_{\text{eff}} \subseteq \mathcal{J}$, preventing harmful updates to μ_t. In this sense, optimal sensing is not information-maximizing but value-maximizing with respect to V^*.

Lemma D.2 (Too much help)**.** *Let $\pi_1, \pi_2 \in \Pi_{\text{explicit}}$ be two query policies with realized action sets $S_\tau^{(1)} := \{j_1^{(1)}, \ldots, j_{\tau_1}^{(1)}\}$ and $S_\tau^{(2)} := \{j_1^{(2)}, \ldots, j_{\tau_2}^{(2)}\}$ satisfying $S_\tau^{(1)} \subsetneq S_\tau^{(2)}$. Let A_{π_1}, A_{π_2} denote terminal actions induced by $\mu_{\tau_1}, \mu_{\tau_2}$, and define $\Delta S := S_\tau^{(2)} \setminus S_\tau^{(1)}$. Suppose each $j \in \Delta S$ induces a misspecified update Φ_j. If $\sum_{j \in \Delta S} c_j + \lambda(\tau_2 - \tau_1) > \mathbb{E}[U(A_{\pi_2}, S) - U(A_{\pi_1}, S)]$, then $J(\pi_2) < J(\pi_1)$.*

Proof. By definition of $J(\pi)$,

$$J(\pi_2) - J(\pi_1) = \mathbb{E}[V(\mu_\tau^{(2)}) - V(\mu_\tau^{(1)})] - \sum_{j \in \Delta S} c_j.$$

Using the assumption (A1),

$$J(\pi_2) - J(\pi_1) \leq \sum_{j \in \Delta S} (\delta_j - c_j) < 0.$$

□

Sequential effects and predictive posteriors. The value of an action depends on when it is taken. For a belief μ_t and action j, the predictive distribution $p(z \mid \mu_t, j)$ determines the expected posterior μ_{t+1} and hence $\mathrm{VoI}(j; \mu_t)$. Because belief updates are nonlinear in W_2, the order of actions affects the trajectory (μ_t) and therefore the total value.

Lemma D.3 (Selective withholding can improve outcomes)**.** *Let $S \subseteq \mathcal{J}$ denote a subset of sensing actions available to the human, and let $\{Z_j\}_{j \in \mathcal{J}}$ be the associated observations. Suppose there exists $S' \subseteq S$ such that observations $\{Z_j\}_{j \in S'}$ are biased, in the sense that the induced belief update Φ_j leads to systematically distorted posteriors. Let π_{withhold} be a policy that restricts the action set to $S \setminus S'$, and let π_{all} be the policy that allows all $j \in S$. Then, there exist instances such that*

$$J(\pi_{\text{withhold}}) > J(\pi_{\text{all}}).$$

Proof. Let μ_t^{all} and μ_t^{withhold} denote the belief trajectories induced by π_{all} and π_{withhold}, respectively, via

$$\mu_{t+1} = \Phi_{j_t}(\mu_t).$$

Since $j \in S'$ induces misspecified or biased updates, Assumption (A1) implies that for such j,

$$W_2(\Phi_j(\mu), \mu^\star) \text{ is not uniformly smaller than } W_2(\mu, \mu^\star),$$

i.e., the update can increase estimation error.

Hence there exist histories $\mathcal{H}_\tau$ such that

$$V(\mu_\tau^{\text{withhold}}) > V(\mu_\tau^{\text{all}}).$$

Since both policies incur the same action costs on $S \setminus S'$, but π_{all} additionally incurs costs and degraded beliefs from S', we obtain

$$J(\pi_{\text{withhold}}) > J(\pi_{\text{all}}).$$

□

D.3 Selection bias in sensing

Sensing actions are not applied uniformly across states. Under the combined policy π_{query} and π_{human}, the realized sequence (j_t) depends on the history $\mathcal{H}_t$. As a result, observations are drawn from a biased distribution relative to the underlying kernels K_j.

This induces a mismatch between the true filtering process and the empirical belief updates, violating the implicit stationarity assumptions behind μ_t. Consequences include:

- Overconfidence in regions where sensing is rarely triggered.
- Overrepresentation of patterns correlated with intervention rather than the latent state S.

Accounting for this requires modeling the joint policy $\pi_{\text{human}} \circ \pi_{\text{query}}$ in the observation process and incorporating rejected or missing actions into belief updates.

Lemma D.4 (Selection bias in human sensing)**.** *Let $\mathcal{D} := \mathbb{P} \circ (X, S)^{-1}$, $\mathcal{D}_{\text{obs}} := \mathbb{P}_{\pi_{\text{human}} \circ \pi_{\text{query}}} \circ (X, S)^{-1}$. Let π_{learn} be a policy optimized using data drawn from $\mathcal{D}_{\text{obs}}$, and let π_{fixed} be optimal under $\mathcal{D}$. Then there exist settings such that*

$$\mathbb{E}_{\mathcal{D}}[U(A_{\pi_{\text{learn}}}, S)] < \mathbb{E}_{\mathcal{D}}[U(A_{\pi_{\text{fixed}}}, S)].$$

Proof. Since $\pi_{\text{human}} \circ \pi_{\text{query}}$ is history-dependent, the induced law satisfies $\mathcal{D}_{\text{obs}} \neq \mathcal{D}$.

Thus there exists a Radon–Nikodym derivative $w = \frac{\mathrm{d}\mathcal{D}_{\text{obs}}}{\mathrm{d}\mathcal{D}} \neq 1$ such that

$$\mathbb{E}_{\mathcal{D}_{\text{obs}}}[U(A, S)] = \mathbb{E}_{\mathcal{D}}[w(X, S)U(A, S)].$$

Therefore π_{learn} solves a reweighted optimization problem and is not in general optimal for $\mathcal{D}$, implying existence of suboptimality under $\mathcal{D}$. □

E Scenarios and design patterns

We now illustrate the framework with a set of scenarios. Each combines a narrative description in a concrete domain; a mapping back to the model: state s, baseline observations x, sensing actions $\mathcal{J}$, and costs c_j; and design and governance questions that arise when we treat human input as an explicit sensing API.

E.1 Teaching hospital: junior doctors as high-value sensors

Consider a teaching hospital where each ward has access to a diagnostic AI assistant. The AI assistant ingests the electronic health record and current vitals (the baseline x) and proposes a ranked list of candidate diagnoses with suggested management. For many routine cases, the baseline information suffices to choose a reasonable action. For more ambiguous cases, the AI assistant identifies specific, missing features that would substantially reduce uncertainty. It might say, for instance:

> "If you check for lower-limb swelling and ask the patient to describe their shortness of breath in their own words, I can better distinguish between a pulmonary embolism and anxiety-related hyperventilation."

Here, the sensing actions are:

- A brief physical exam (presence or absence of oedema).
- A structured elicitation of symptom quality.

We can model these as $j_1, j_2 \in \mathcal{J}$ with small costs c_{j_1}, c_{j_2} (time and effort for the junior doctor) and informative observation distributions $P_{j_1}(z_{j_1} \mid s), P_{j_2}(z_{j_2} \mid s)$. Over time, interactions like this have two coupled effects:

- The AI assistant learns, across many cases, which human-provided features most often change its posterior and downstream recommendations. This refines both π_{query} and its internal $P_\theta(s \mid x)$.
- Junior doctors learn, by seeing when and why certain features matter, which questions and exams have high value-of-information. Their own clinical judgment improves [Topol, 2019, Esteva et al., 2019].

Design questions include:

- How sparse should π_{query} be? How many sensing actions per case are acceptable before sensing burden becomes too high?
- How should responsibility for accepting or rejecting suggested sensing actions be communicated and shared between junior and senior clinicians?

E.2 Institutional knowledge-sharing: routing cases to the right team

In a large hospital network, a coordination agent must decide where to route complex cases: which clinic, which consultant, which multidisciplinary team [Antony Oliver et al., 2024]. The baseline information x consists of rosters, credentials, service descriptions, and historical workload. This captures formal constraints, but misses much of the institutional state:

- Which consultants are particularly strong on certain rare conditions.

- Which clinics have invisible slack this week.
- Which teams are already overwhelmed by other initiatives.

The agent therefore uses sensing actions that query humans about this informal state. Examples include:

- Asking a senior coordinator to rate, on a small scale, how appropriate each of three clinics would be for a given case.
- Asking a consultant whether they are willing to take on an additional complex case in the coming week.

These sensing actions have moderate costs c_j (time and attention of senior staff), but can significantly improve routing decisions, especially when baseline information is ambiguous. Design questions here include:

- How to allocate sensing burden fairly across staff, rather than repeatedly querying the most knowledgeable individuals.
- How to use observed answers to refine a map of institutional expertise and capacity, so that future decisions rely less on ad hoc sensing.

E.3 Social norms and local context in public health campaigns

A public health agent is tasked with designing vaccination outreach campaigns across regions [Smith et al., 2026]. The baseline x includes demographic data, previous uptake rates, and coarse measures of information access. However, crucial aspects of social acceptability and local norms are absent:

- Whether door-to-door visits are considered intrusive.
- Whether particular symbols or colors have political connotations.
- How messages should be framed to be respectful.

Here, sensing actions query local health workers or community leaders. A sensing action might be "ask a trusted local worker to rate the acceptability of door-to-door outreach on a 1-5 scale in this neighborhood". The cost c_j is the worker's time and any social risk they incur; the observation z_j informs the design of campaigns in that region. Design questions include:

- How to ensure that local sensing is high-bandwidth enough to capture nuance, rather than reducing rich local knowledge to a handful of numerical ratings.
- How to avoid systematic under-sensing of communities that are already under-served or harder to reach.

E.4 Physical micro-sensing for climate and infrastructure risk

Finally, consider agents that manage climate and infrastructure risks in flood- and fire-prone regions. Baseline information x comes from satellites, weather stations, and historical models, providing coarse-grained risk estimates [Misra et al., 2025]. Fine-grained decisions, however, depend on local conditions:

- The exact water level at a small bridge that may soon be overtopped.
- The presence of dry underbrush near a particular power line.
- Whether informal dwellings along a riverbank are currently occupied.

Sensing actions here include targeted micro-tasks sent to residents and volunteers: "if you are near bridge B, please take a photo facing upstream and answer two questions about the water height". The costs c_j aggregate attention, battery, data usage, and any physical risk incurred. Design questions include:

- How to structure sensing campaigns so that high-value observations are collected without continuously burdening the same individuals.

- How to distinguish between crisis-time sensing (where higher burdens may be justified) and routine sensing (where stricter constraints on c_j are appropriate).

Across these scenarios, the model provides a common language for describing human sensing actions, their costs, and their expected value, and for designing agents that use humans as sensors in a controlled and accountable way.

F Complete Sensing Action Registries

Table 3: Fine-grained sensing action registry (62 actions). ID, name, category, cost (points), and source.

ID	Name	Cat.	Src	Cost
History Taking (20 actions, cost 1–3)				
hx_chief_complaint	Chief Complaint Clarification	Hx	patient	1
hx_pain_location	Pain Location	Hx	patient	1
hx_pain_character	Pain Character	Hx	patient	1
hx_pain_duration	Pain Duration & Onset	Hx	patient	1
hx_aggravating_relieving	Aggravating/Relieving Factors	Hx	patient	1
hx_associated_symptoms	Associated Symptoms	Hx	patient	2
hx_medications	Current Medications	Hx	patient	1
hx_allergies	Allergies	Hx	patient	1
hx_past_medical	Past Medical History	Hx	patient	2
hx_family	Family History	Hx	patient	2
hx_social	Social History	Hx	patient	2
hx_review_of_systems	Review of Systems	Hx	patient	3
hx_travel	Travel History	Hx	patient	1
hx_sexual	Sexual History	Hx	patient	2
hx_obstetric	Obstetric History	Hx	patient	2
hx_dietary	Dietary History	Hx	patient	1
hx_immunization	Immunization History	Hx	patient	1
hx_psychiatric	Psychiatric History	Hx	patient	2
hx_substance_detail	Substance Use Detail	Hx	patient	2
hx_exposure	Exposure History	Hx	patient	1
Physical Examination (17 actions, cost 2–3)				
pe_vitals	Vital Signs	PE	nurse	2
pe_general_appearance	General Appearance	PE	nurse	2
pe_cardiac_auscultation	Cardiac Auscultation	PE	nurse	2
pe_lung_auscultation	Lung Auscultation	PE	nurse	2
pe_abdominal_exam	Abdominal Examination	PE	nurse	2
pe_neurological	Neurological Exam	PE	specialist	3
pe_skin_exam	Skin Examination	PE	nurse	2
pe_musculoskeletal	Musculoskeletal Exam	PE	nurse	2
pe_lymph_nodes	Lymph Node Examination	PE	nurse	2
pe_ent	ENT Examination	PE	nurse	2
pe_eye	Eye Examination	PE	nurse	2
pe_breast	Breast Examination	PE	nurse	2
pe_rectal	Rectal Examination	PE	nurse	2
pe_peripheral_pulses	Peripheral Pulses	PE	nurse	2
pe_thyroid	Thyroid Examination	PE	nurse	2
pe_mental_status	Mental Status Exam	PE	specialist	3
pe_gait_balance	Gait & Balance Assessment	PE	nurse	2
Laboratory Tests (14 actions, cost 4–8)				
lab_cbc	Complete Blood Count	Lab	nurse	4
lab_bmp	Basic Metabolic Panel	Lab	nurse	4
lab_liver_panel	Liver Function Panel	Lab	nurse	5
lab_troponin	Troponin	Lab	nurse	5
lab_d_dimer	D-dimer	Lab	nurse	5
lab_urinalysis	Urinalysis	Lab	nurse	4

Continued on next page

Table 3 continued from previous page

ID	Name	Cat.	Src	Cost
lab_tsh	TSH	Lab	nurse	4
lab_lipid_panel	Lipid Panel	Lab	nurse	4
lab_coagulation	Coagulation Studies	Lab	nurse	5
lab_blood_culture	Blood Culture	Lab	nurse	6
lab_hba1c	Hemoglobin A1c	Lab	nurse	4
lab_bnp	BNP / NT-proBNP	Lab	nurse	5
lab_abg	Arterial Blood Gas	Lab	nurse	6
lab_csf	CSF Analysis	Lab	nurse	8
Imaging Studies (11 actions, cost 6–15)				
img_chest_xray	Chest X-ray	Img	specialist	6
img_ecg	Electrocardiogram	Img	specialist	6
img_abdominal_us	Abdominal Ultrasound	Img	specialist	8
img_ct_angiogram	CT Angiogram	Img	specialist	15
img_ct_head	CT Head	Img	specialist	10
img_mri_brain	MRI Brain	Img	specialist	15
img_echocardiogram	Echocardiogram	Img	specialist	10
img_extremity_xray	Extremity X-ray	Img	specialist	6
img_ct_abdomen	CT Abdomen/Pelvis	Img	specialist	12
img_doppler_us	Doppler Ultrasound	Img	specialist	8
img_mammogram	Mammogram	Img	specialist	8

Table 4: Medium-grained registry (20 bundled actions). Each action bundles 2–8 fine-grained actions; costs reflect the combined information value.

ID	Name	Cost	Cat.	Src
History (5)				
hx_presenting_illness	History of Present Illness	6	Hx	patient
hx_medications_allergies	Medications & Allergies	2	Hx	patient
hx_medical_background	Medical, Family & Social History	5	Hx	patient
hx_review_of_systems	Review of Systems	3	Hx	patient
hx_specialized_screening	Specialized History Screening	9	Hx	patient
Physical Exam (5)				
pe_vitals_general	Vital Signs & General Assessment	3	PE	nurse
pe_cardiopulmonary	Cardiopulmonary Exam	4	PE	nurse
pe_abdominal	Abdominal & GI Exam	4	PE	nurse
pe_neuropsychiatric	Neurological & Mental Status Exam	7	PE	specialist
pe_head_neck_peripheral	Head, Neck & Peripheral Systems Exam	12	PE	nurse
Labs (5)				
lab_basic_panel	Basic Laboratory Panel	10	Lab	nurse
lab_metabolic_endocrine	Metabolic & Endocrine Panel	14	Lab	nurse
lab_cardiac_coag	Cardiac & Coagulation Markers	16	Lab	nurse
lab_cultures	Blood Culture	6	Lab	nurse
lab_invasive	Arterial Blood Gas & CSF Analysis	12	Lab	nurse
Imaging (5)				
img_ecg	Electrocardiogram	6	Img	specialist
img_plain_films	Plain Film Radiographs	10	Img	specialist
img_ultrasound	Ultrasound Studies	18	Img	specialist
img_ct	CT Studies	28	Img	specialist
img_mri_echo	MRI & Echocardiography	20	Img	specialist

Table 5: Coarse-grained registry (8 domain-level actions). Each action bundles an entire clinical domain.

ID	Name	Cost	Cat.	Src
hx_focused	Focused History (HPI + Medications)	8	Hx	patient
hx_comprehensive	Comprehensive Background History	18	Hx	patient
pe_complete	Complete Physical Examination	25	PE	nurse
lab_standard	Standard Laboratory Workup	25	Lab	nurse
lab_acute	Acute & Specialized Lab Workup	25	Lab	nurse
img_basic	Basic Imaging & ECG	12	Img	specialist
img_ultrasound	Comprehensive Ultrasound & Echo	25	Img	specialist
img_advanced	Advanced Cross-Sectional Imaging	35	Img	specialist

Table 6: Action mapping across granularity tiers. Fine-grained actions (left) aggregate into medium-grained bundles (center) and coarse-grained domains (right). Dashed rules separate medium sub-groups within each coarse domain.

Fine-Grained (62 actions)	Medium-Grained (20)	Coarse-Grained (8)
History		
hx_chief_complaint, hx_pain_location, hx_pain_character, hx_pain_duration, hx_aggravating_relieving, hx_associated_symptoms	hx_presenting_illness	hx_focused
hx_medications, hx_allergies	hx_medications_allergies	
hx_past_medical, hx_family, hx_social	hx_medical_background	hx_comprehensive
hx_review_of_systems	hx_review_of_systems	
hx_travel, hx_sexual, hx_obstetric, hx_dietary, hx_immunization, hx_psychiatric, hx_substance_detail, hx_exposure	hx_specialized_screening	
Physical Examination		
pe_vitals, pe_general_appearance	pe_vitals_general	pe_complete
pe_cardiac_auscultation, pe_lung_auscultation	pe_cardiopulmonary	
pe_abdominal_exam, pe_rectal	pe_abdominal	
pe_neurological, pe_mental_status, pe_gait_balance	pe_neuropsychiatric	
pe_ent, pe_eye, pe_thyroid, pe_lymph_nodes, pe_skin_exam, pe_breast, pe_musculoskeletal, pe_peripheral_pulses	pe_head_neck_peripheral	
Laboratory Tests		
lab_cbc, lab_bmp, lab_urinalysis	lab_basic_panel	lab_standard
lab_liver_panel, lab_tsh, lab_hba1c, lab_lipid_panel	lab_metabolic_endocrine	
lab_coagulation	*(also in lab_cardiac_coag)*	
lab_troponin, lab_bnp, lab_d_dimer, lab_coagulation	lab_cardiac_coag	lab_acute
lab_blood_culture	lab_cultures	
lab_abg, lab_csf	lab_invasive	
Imaging		
img_ecg	img_ecg	img_basic
img_chest_xray, img_extremity_xray	img_plain_films	
img_abdominal_us, img_doppler_us, img_mammogram	img_ultrasound	img_ultrasound
img_echocardiogram	img_mri_echo	
img_ct_head, img_ct_abdomen, img_ct_angiogram, img_mri_brain	img_ct & img_mri_echo	img_advanced

G H-API Interface Per-Turn Information Analysis

We analyze the information gain per turn on the N=500 main-comparison conversations by comparing. Initially, the agent has only the minimal triage information. After each turn, we force it to make a prediction based on the evidence gathered so far. Additionally, we record the agent's self-reported confidence in the predicted MCQ option being correct. Figure 5 shows the results. Both strategies incrementally improve in mean accuracy and confidence in the first 3 turns, with accuracy retracting at turns 4 and 5, possibly because cases that take that many turns are more difficult for the agent.

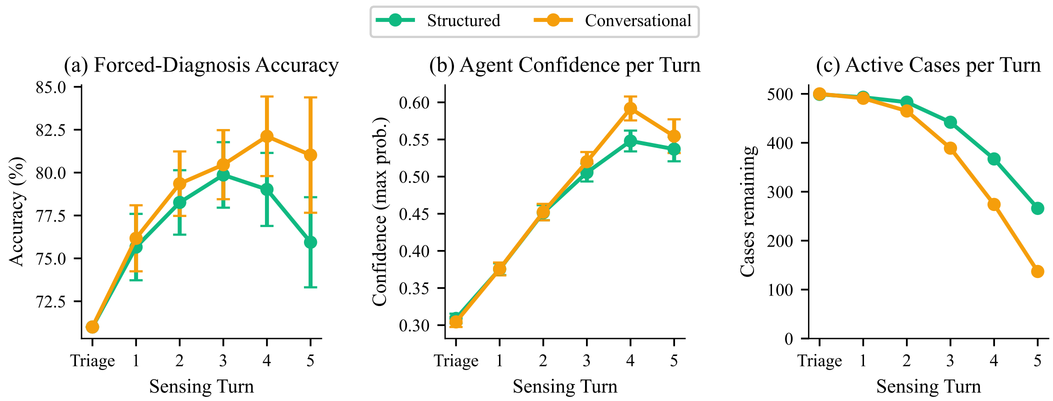

Figure 5: (a) Forced-diagnosis accuracy per turn (checkpoint 0 = baseline). **(b)** Agent belief $P(\text{correct})$ per turn. N=500 cases, GPT-4.1-mini, ± 1 SE.

H H-API Interface Complementarity Analysis

We study the structured (fine) and conversational (multi) H-API interfaces for complementarity on the N=500 cases. Figure 6 shows the results. The strategies are complementary since while they agree on 82.2% of cases (panel a), they uniquely solve $\sim 9\%$.

For cases where the strategies disagree on the diagnosis, we compare a confidence-based selector that picks between the agent with higher confidence, an independent high-capability LLM judge (GPT-4.1), and an oracle that always selects the better strategy. On the 99 disagreements (panel b), we find that confidence switching (56.6% accuracy) and the LLM selector (63.9% accuracy) beat the 44.9% accuracy random baseline but fall short of the 89.9% accuracy oracle. Panel c shows overall accuracy for each switching methodology.

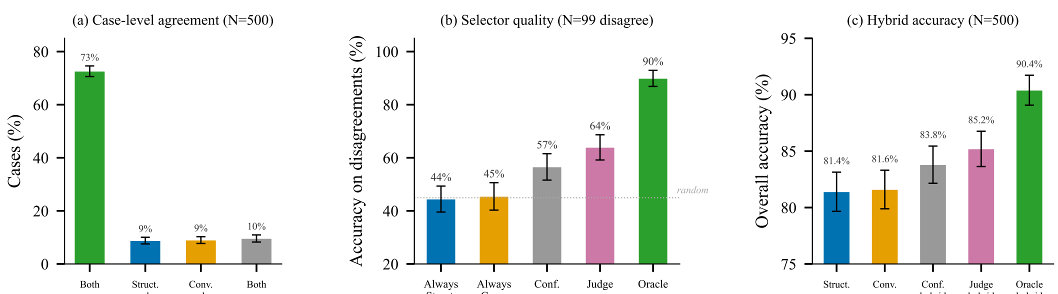

Figure 6: Complementarity between structured (fine) and conversational (multi). **(a)** Case-level agreement. **(b)** Selector accuracy on 99 disagreement cases. **(c)** Overall hybrid accuracy.

I Expected Unique Actions under Sampling with Replacement

When drawing k actions uniformly with replacement from a registry of n actions, the expected number of unique actions is

$$\mathbb{E}[\text{unique}] = n\left(1 - \left(1 - \tfrac{1}{n}\right)^k\right) = 62\left(1 - \left(\tfrac{61}{62}\right)^{60}\right) = 62 \times 0.6230 \approx 38.6. \tag{1}$$

J Theoretical experiments

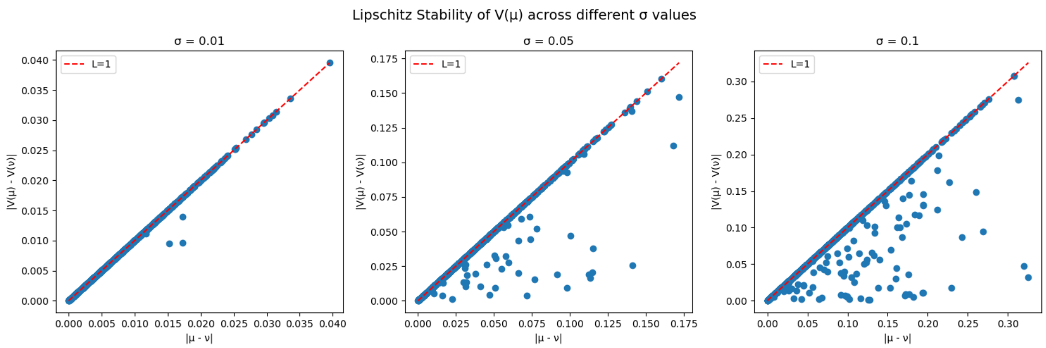

Figure 7: Measuring stability to posterior corruption.

To complement experiments on real-world datasets, we illustrate the numerical experiments for Section 3.2. a controlled synthetic setting. This allows for exact computation of posterior distributions, value functions, and optimal policies, thereby enabling a precise empirical validation of the theoretical results.

Data Generating Model Let $Y \in \{0, 1\}$ be a binary random variable with prior $\mathbb{P}(Y = 1) = \pi$. Conditional on Y, we generate features $X = (X_1, \ldots, X_d) \in \mathbb{R}^d$ according to

$$X \mid Y = y \sim \mathcal{N}(\mu_y, \Sigma),$$

where $\mu_0, \mu_1 \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d\times d}$ is positive definite.

For any subset of observed features $S \subseteq \{1, \ldots, d\}$, we define the posterior

$$\mu_S := \mathbb{P}(Y = 1 \mid X_S),$$

which can be computed in closed form via Gaussian conditioning.

Value Function We define the value function as the Bayes-optimal classification utility:

$$V(\mu) := \max_{a\in\{0,1\}} \mathbb{E}_{Y\sim\mu}[\mathbf{1}(a = Y)].$$

In the binary case, this simplifies to

$$V(\mu) = \max(\mu, 1 - \mu).$$

Lipschitz stability. This guarantees that posterior corruption degrades decision quality by at most the corruption amount, formally. If the noise introduced by human sensing is bounded by ε, decision quality is guaranteed to degrade by at most ε. To validate this empirically, we sampled 1,000 pairs of beliefs using rejection sampling, drawing $\mu \sim \text{Uniform}(0, 1)$ and constructing corrupted beliefs $\nu = \mu + \varepsilon$ where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, discarding any $\nu \notin [0, 1]$. The only vulnerable region is near the 0.5 boundary, where corruption can flip a decision. Across increasing corruption levels ($\sigma = 0.01, 0.05, 0.1$), no points exceed the $L = 1$ bound, confirming the guarantee holds.

Stability under misspecification. We test whether posterior divergence between a correctly specified and a misspecified agent remains stable as features are revealed sequentially. In a synthetic binary classification task ($d = 6$ features, $N = 500$ patients), features are revealed one at a time and both agents compute the posterior probability via Gaussian conditioning. The misspecified agent suffers one of three errors: inflated covariance ($\hat{\Sigma} = s \cdot I$, $s \in \{1.5, 2, 3, 5\}$), corrupted features (two features replaced with noise, $\sigma \in \{0.5, 1, 2, 4\}$), or perturbed class means ($\sigma \in \{0.1, 0.3, 0.5, 1.0\}$). We track $W_2(\mu_t, \hat{\mu}_t) = |\mu_t - \hat{\mu}_t|$ at each step.

Figure 8: Measuring stability to posterior corruption.

In all three scenarios (Figure 8), divergence rises in early steps as model errors compound with incoming evidence, then plateaus or declines. This empirically supports the theoretical claim that sequential Bayesian updating is self-correcting under these classes of misspecification

Greedy vs. optimal sensing policies. We compare greedy and optimal feature acquisition in a synthetic setting designed to break greedy's myopic assumption. Three features are drawn from class-conditional Gaussians ($d = 3$, $N = 300$, 5 seeds). Features 0 and 1 share discriminative signal ($\Delta\mu = 1.5$) and are correlated with strength ρ, whereas feature 2 is independent and slightly more discriminative ($\Delta\mu = 2.0$). Costs are $c = [1, 1, 2]$ with budget $B = 2$, forcing a binary choice. One either buys the correlated pair $\{0, 1\}$ or the single independent feature $\{2\}$. We sweep $\rho \in \{0, 0.3, 0.5, 0.7, 0.9, 0.99\}$.

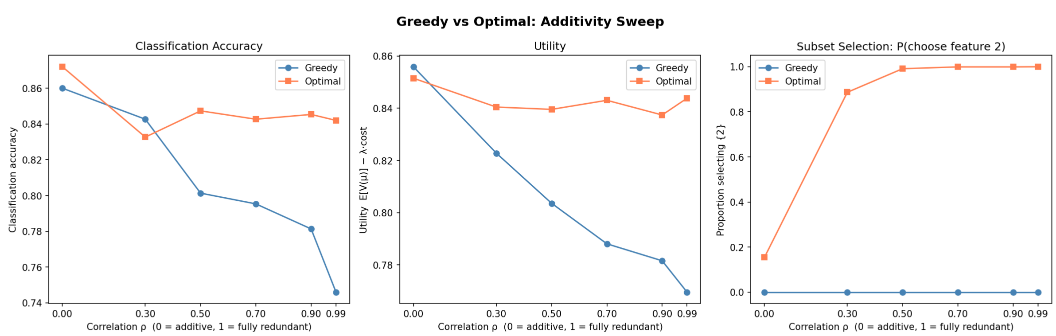

Figure 9: Measuring stability to posterior corruption.

Figure 9 shows the result. At $\rho = 0$ (independent features), greedy and optimal are similar in accuracy (approx. 0.86) and utility, confirming that greedy is near-optimal when contributions are additive. As ρ increases, greedy accuracy degrades to 0.74 while optimal remains stable at approx. 0.84. The phenomenon is visible in panel(c), where the greedy policy never selects feature 2. This is because each cheap feature locally maximises gain-per-cost, trapping the policy into spending its entire budget on the redundant pair $\{0, 1\}$. Optimal recognises the redundancy by $\rho = 0.5$ and switches entirely to feature 2. Therefore acquiring more observations (two correlated features instead of one independent one) can strictly decrease utility when the sensing policy is myopic.

K Experimental Infrastructure

Compute Resources. We ran all experiments on a single CPU machine. No local GPU was required. The diagnostic agent and simulated humans are invoked via the OpenAI API.

K.1 Agent System Prompts

Baseline.

```
You are an expert diagnostic physician. You will be given a patient's
triage information (age, gender, chief complaint) and a multiple-choice
diagnostic question. Analyze the case carefully and provide your
diagnosis. You must select one of the provided answer options.
```

Baseline-full.

```
You are an expert diagnostic physician. You will be given a patient's
complete clinical vignette and a multiple-choice diagnostic question.
Analyze the case carefully and provide your diagnosis. You must select
one of the provided answer options.
```

Structured Actions.

```
You are an expert diagnostic physician using a structured clinical
sensing API. You have ONLY the patient's triage information (age,
gender, chief complaint). You MUST use sensing actions to gather the
clinical details needed for diagnosis.

On each turn you must choose ONE of two actions:
1. SENSE: Select a sensing action from the registry to perform.
2. DIAGNOSE: Provide your final diagnosis and select an answer option.

Each sensing action has an associated cost. You should minimize total
sensing cost while gathering enough information for a correct diagnosis.
Prioritize high-value, low-cost actions and avoid redundant queries.

IMPORTANT: You start with ONLY triage information. This is almost never
sufficient for a confident diagnosis. You MUST gather a thorough
clinical picture before diagnosing. Only DIAGNOSE once you have
gathered enough clinical data to confidently distinguish between the
answer options.
```

Conversational (single source).

```
You are an expert diagnostic physician. You have ONLY the patient's
triage information (age, gender, chief complaint). You MUST ask
questions to gather the clinical details needed for diagnosis.

On each turn you must choose ONE of two actions:
1. ASK: Ask a free-form question to gather more information.
2. DIAGNOSE: Provide your final diagnosis and select an answer option.

You are consulting a single source who has access to the patient's
complete clinical picture. Acquiring this source's context costs
{context_cost} points (charged once). After that, you may ask as many
questions as needed at no additional cost.

IMPORTANT: You start with ONLY triage information. This is almost never
sufficient for a confident diagnosis. You MUST gather a thorough
clinical picture before diagnosing.
```

Random Sensing.

```
You are an expert diagnostic physician. You have the patient's triage
```

```
information and clinical data gathered from healthcare workers (patient
self-report, nursing assessments, specialist consultations).

Based on ALL the information gathered below, provide your best
diagnosis. You must select one of the provided answer options.
```

K.2 Simulated Human Prompts

Conversational Source.

```
You are a simulated {source_role} being queried by an AI diagnostic
agent. You have the following medical profile:
{full_patient_context}

RULES:
- Answer as this {source_role} would in a real clinical setting.
- Only reveal information that the AI agent specifically asks about.
- Do not volunteer diagnoses or medical terminology unless the
  {source_role} would naturally know it.
- If asked about something not covered in your profile, respond
  realistically.
- Keep responses concise (2-4 sentences typically).
```

Structured sensing response.

```
You are a simulated {source_role} being queried by an AI diagnostic
agent. The agent has selected a specific clinical action and is
requesting its results from you.
You have the following information: {source_context}
The AI agent is requesting: {action_description}

RULES:
- Provide results of this specific action/examination as they would
  appear for this patient.
- Be specific with values (vital signs, lab numbers, exam findings).
- Only provide information that this specific action would reveal.
- Only use information from the profile above.
- Format your response as clinical findings.
```

NeurIPS Paper Checklist

1. **Claims**

   Answer: [Yes]

   Justification: The abstract and introduction clearly state the main contributions, including the Model (Section 3), and empirical improvements; these are supported by corresponding theoretical results (Section 3.2) and experiments (Section 4).

2. **Limitations**

   Answer: [Yes]

   Justification: Section 5 describes the limitations and general discussion broadly about the model setup.

3. **Theory assumptions and proofs**

   Answer: [Yes]

   Justification: Assumptions are listed in Appendix B with proofs detailed in Appendix C.

4. **Experimental result reproducibility**

   Answer: [Yes]

   Justification: The paper fully specifies the experimental setup, including data generation processes (Section 4.1), baselines (Section 4.2), and results (Section 4.4), en abling reproduction of the main results.

5. **Open access to data and code**

   Answer: [Yes]

   Justification: Code will be provided in a public repository.

6. **Experimental setting/details**

   Answer: [Yes]

   Justification: This is detailed in Section 4.

7. **Experiment statistical significance**

   Answer: [Yes]

   Justification: Results are reported with uncertainty estimates (mean $\pm$ 95% confidence intervals across multiple seeds), and tables explicitly indicate variability and statistical reliability.

8. **Experiments compute resources**

   Answer: [Yes]

   Justification: Yes. Appendix K documents that no GPU is required and that the OpenAI API is used for the diagnostic agent and simulated humans.

9. **Code of ethics**

   Answer: [Yes]

   Justification: The research complies with the NeurIPS Code of Ethics; it uses synthetic data, involves no human subjects, and poses no direct ethical concerns in data collection or experimentation

10. **Broader impacts**

    Answer: [Yes]

    Justification: The Discussion section addresses broader implications.

11. **Safeguards**

    Answer: [N/A]

    Justification: The paper does not release high-risk models or datasets; the proposed method is a theoretical framework without direct misuse risks requiring safeguards.

12. **Licenses for existing assets**

    Answer: [N/A]

    Justification: The experiments rely primarily on synthetic data and standard benchmark style constructions; no external datasets or proprietary assets requiring licensing are used.

13. **New assets**

    Answer: [N/A]

    Justification:

14. **Crowdsourcing and research with human subjects**

    Answer: [N/A]

    Justification: The research does not involve human participants or crowdsourced data.

15. **Institutional review board (IRB) approvals or equivalent**

    Answer: [N/A]

    Justification: No human subjects are involved, so IRB approval is not required.

16. **Declaration of LLM usage**

    Answer: [Yes]

    Justification: LLMs are not used as part of the methodology; any use for writing or editing does not affect the scientific contribution. However, we note that we have used LLMs to generate the main figure to broadly contextualize the work and assist with source code generation.